



Hierarchical Cross-Modality Interaction for Unified Video-Text Retrieval Modeling

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Abstract. Existing video-text retrieval (VTR) methods suffer from two primary limitations: they typically extract features from each modality in isolation and struggle to effectively align information across different granularity levels. These issues neglect potential contextual cues between the two modalities and can lead to a trade-off between capturing global context and fine-grained details. To address these challenges, we introduce the Hierarchical Cross-Modality Interaction (HCMI) framework, a novel approach that performs bidirectional, multi-level feature interaction during training. Our framework systematically aligns features at patch-word, frame-sentence, and global video-sentence levels, leveraging a dual attentive module for deep feature enhancement and a cross-modality adaptor for efficient feature distillation. Crucially, HCMI requires only a single modality's input during inference, making it highly practical for large-scale retrieval tasks. Our method achieves new state-of-the-art results on several challenging benchmarks, including MSR-VTT, DiDeMo, VATEX, MSVD and Activity-Net, demonstrating its significant advantages.

Keywords: Bidirectional VTR · Cross Attention · Feature Alignment

1 Introduction

Video-Text Retrieval (VTR), a fundamental task in cross-modal understanding, aims to bridge the semantic gap between visual and textual data [1, 12, 29]. Early approaches focused on learning a joint embedding space where corresponding video and text pairs are projected closely [20, 28]. More recently, the advent of powerful vision-language pre-trained models, particularly CLIP [19], has significantly advanced the state-of-the-art. Many leading methods now adapt CLIP's image and text encoders for the video domain [15–17], typically by adding temporal modeling modules to aggregate frame-level features into a global video representation before computing cross-modal similarity [6, 16]. To capture more nuanced alignments, subsequent works have explored finer-grained matching, such as aligning individual frames with sentences [7, 15] or even video patches with words [32].

Despite these advancements, existing methods suffer from two primary limitations. First, they typically extract features from a single source modality in isolation. The video encoder processes visual content without any knowledge of the text query it will be compared against, and vice versa. This approach neglects the potential of using the target modality’s information as a contextual cue to guide the extraction of more discriminative and relevant features. For instance, given the query “a person slicing a cucumber”, an ideal video encoder would amplify features related to the specific action of “slicing” and the object “cucumber”, while suppressing irrelevant background details. Second, current models often struggle to effectively align features across different levels of granularity. Methods that operate at a global level (video-to-sentence) may capture overall context but miss crucial fine-grained details, while patch-to-word alignment excels at details but can lose the broader semantic context. This creates a need for a unified framework that can coherently integrate multi-level correspondences.

To address these challenges, we introduce the Hierarchical Cross-Modality Interaction (HCMI) framework, a novel approach for unified video-text retrieval. Our key idea is to enable multi-level, bidirectional interaction between modalities during the feature encoding process itself, while still allowing for efficient single-modality inference. HCMI systematically addresses the density and granularity mismatch between video and text by performing hierarchical alignment at the fine-grained (patch-word), coarse-grained (frame-sentence), and global (video-sentence) levels. This is achieved through a two-stage training paradigm where cross-modal interaction enriches the representations. Crucially, during inference, the model only requires a single modality’s input to generate these enhanced features. To summarize, our main contributions are as follows:

- We propose the Hierarchical Cross-Modality Interaction (HCMI) framework, a novel paradigm that performs bidirectional, multi-level feature interaction during training yet requires only a single modality for efficient inference, resolving the common trade-off between performance and retrieval speed.
- We design a Dual Attentive Module (DAM) for deep bidirectional feature enhancement, along with a Cross-Modality Adaptor (CMA) for single-modality feature distillation, effectively integrating hierarchical information to balance fine-grained details with global context.
- We achieve new state-of-the-art results on several challenging benchmarks, including MSR-VTT, DiDeMo, VATEX, MSVD and Activity-Net, demonstrating the significant advantages of our proposed method.

The remainder of this paper is organized as follows. Sect. 2 reviews related work; Sect. 3 details the HCMI architecture, DAM and CMA modules, and the training protocol; Sect. 4 presents experimental results and analyses; and the last section concludes with directions for future work.

2 Related Work

This section reviews three major research directions in video-text retrieval: architectural paradigms balancing efficiency and accuracy, mechanisms for cross-

modal alignment, and recent advances in hierarchical modeling with pre-trained foundation models. These directions highlight the limitations of existing methods and motivate our hierarchical framework.

Architectural Paradigms in Video-Text Retrieval. Video-text retrieval models are typically built on two paradigms: dual-encoders and cross-encoders.

Dual-encoders encode video and text independently into a shared embedding space, enabling efficient large-scale retrieval through pre-computation and nearest-neighbor search [11, 16]. While scalable, their lack of cross-modal interaction restricts fine-grained alignment and reduces accuracy on nuanced tasks.

Cross-encoders jointly process both modalities with deep interaction layers, often Transformers, to capture complex inter-modal dependencies [17]. This improves retrieval accuracy but prohibits pre-computation, making them computationally expensive and unsuitable for large-scale deployment. Hybrid strategies that combine dual-encoder and cross-encoder strengths [11] partly alleviate this trade-off, but scalability versus accuracy remains a core challenge.

Cross-Modal Alignment and Interaction Mechanisms. Retrieval performance critically depends on effective alignment across modalities.

Contrastive learning has become the dominant paradigm, aligning video-text pairs by maximizing similarity for positives and minimizing it for negatives [4, 26]. Despite its success, standard formulations can suffer from biased normalization of similarity distributions. This issue has motivated refinements such as normalized contrastive learning [18].

Attention-based fusion, particularly cross-attention in Transformer architectures, has become the standard mechanism for deep interaction [17, 21]. Cross-attention allows queries to selectively attend to relevant visual or temporal segments, enabling context-aware alignment. However, flat attention often fails to capture multi-level semantics (e.g., matching a sentence to a video while also aligning verb phrases to actions) [5, 8], underscoring the need for structured, hierarchical integration.

Hierarchical Modeling and Pre-Trained Foundation Models. To better capture video complexity, recent works explore hierarchical modeling and the use of large-scale vision-language models (VLMs).

Hierarchical approaches explicitly align features at multiple granularities, such as object-word, action-phrase, and event-sentence levels [13, 24, 30]. These methods improve interpretability and enable multi-level semantic reasoning.

Meanwhile, VLMs like CLIP, which are pre-trained on massive image-text datasets, provide powerful general-purpose embeddings transferable to video-text retrieval [14, 25]. Despite their success, image-based pre-training limits temporal modeling, and such models often excel at global alignment but underperform at fine-grained object-level understanding. Augmenting them with fine-grained modules reintroduces scalability issues, leaving unresolved the central gap: how to achieve robust multi-level alignment without sacrificing efficiency.

3 Method

We propose a hierarchical cross-modality interaction framework to address the granularity mismatch and information imbalance in video-text retrieval. The framework is organized into two stages. In stage I, a cross-attention-based hierarchical interaction module learns robust alignment patterns at multiple levels (patch-word, frame-sentence, and video-text). The resulting cross-modal representations are treated as latent soft labels, which encode enriched semantic correspondences. In stage II, we introduce a label pattern distillation strategy that leverages a lightweight adaptor network to project original CLIP features into the cross-attentive feature space. This process enables each video-text pair to inherit its alignment pattern from stage I and refine its modality-specific features accordingly. By coupling hierarchical interaction with knowledge distillation using a single source, our method produces more discriminative and generalizable retrieval features and achieving superior video-text matching performance.

3.1 Problem Definition and Motivation

The Video Text Retrieval task aims to retrieve the most relevant videos from a video database given a text query, or conversely, to retrieve the most relevant texts from a text database given a video query.

Formally, let $V = \{v_1, v_2, \dots, v_N\}$ be a set of videos and $T = \{t_1, t_2, \dots, t_M\}$ be a collection of texts. For *Video-to-Text Retrieval*: Given a query video $v_i \in V$, the objective is to find a ranked list $t'_1, t'_2, \dots, t'_k$ (where $t'_j \in T$), such that t'_j are ordered by their relevance to v_i from highest to lowest. This is particularly useful for generating captions or summaries for unannotated videos. For *Text-to-Video Retrieval*: Given a query text $t_j \in T$, the objective is to find a ranked list $v'_1, v'_2, \dots, v'_k$ (where $v'_j \in V$), such that v'_j are ordered by their relevance to t_j from highest to lowest. This scenario is common in video search engines where users input textual descriptions.

3.2 Stage I: Hierarchical Cross-Modality Interaction for Pattern Label Producing

As illustrated in Fig. 1, Stage I builds a *hierarchical cross-modality interaction* (HCMI) mechanism to mitigate the information-density gap between video and text. We progressively align the two modalities at fine (patch-word), coarse (frame-sentence), and global (video-sentence) granularities via cross-attention, and then fuse them into modality-specific *pattern features*. These fused features act as implicit *soft labels* to supervise Stage II. Since HCMI consumes *both* modalities during inference (which is impractical in retrieval), Stage II adopts an adaptor to distill these interaction patterns so that robust retrieval features can be produced from a *single* modality.

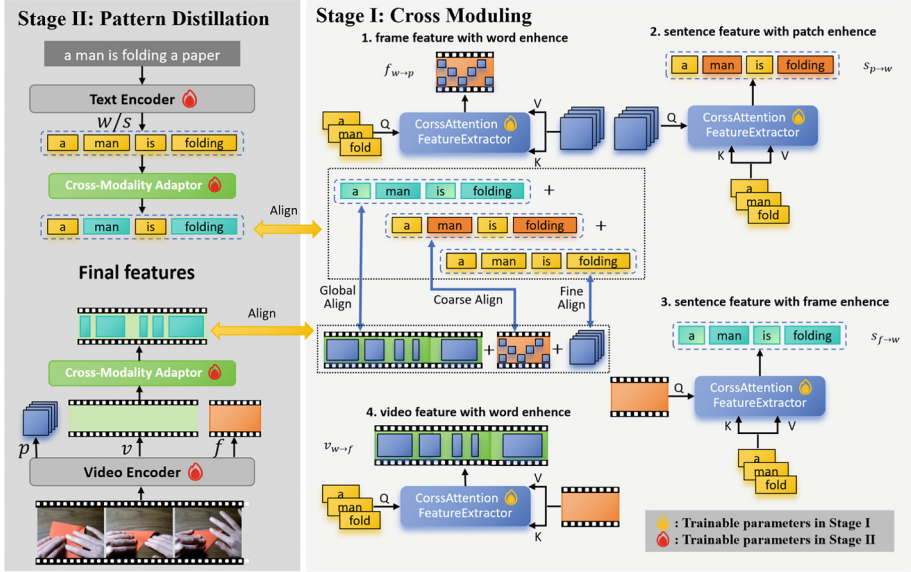


Fig. 1. The main pipeline of the proposed method for multi-modal feature generation.

Original Feature Extraction. We use CLIP encoders for initialization and keep notation compact: the text encoder yields word features w and a sentence feature s ; the video encoder yields patch features p , frame features f , and a global video feature v . These serve as the basis for the subsequent cross-modal interactions.

Fine Alignment Between Original Patch and Word Features. At the most granular level, we align local visual patches with lexical tokens to ground words in concrete regions. We directly use the original features:

$$v_{\text{fine}} = p, \quad t_{\text{fine}} = w \quad (1)$$

Here, cross-attention provides the alignment signal: words query patches (and vice versa) to expose patch–word correspondences that guide higher-level interactions.

Coarse Alignment Between Enhanced Frame and Sentence Features. Building on the fine level, we form frame–sentence interactions enhanced by lower-level cues:

$$v_{\text{coarse}} = \lambda_2 \cdot f_{w \rightarrow p} + f, \quad t_{\text{coarse}} = \alpha_2 \cdot s_{p \rightarrow w} + s, \quad (2)$$

where $f_{w \rightarrow p}$ is obtained by cross-attention with *words as queries* and *patches as keys/values* (word→patch), and $s_{p \rightarrow w}$ is analogously obtained with *patches as queries* and *words as keys/values* (patch→word). This level captures event-centric semantics beyond isolated regions or tokens.

Global Alignment Between Enhanced Video and Sentence Features.

At the highest level, we align the overall video with the whole sentence while injecting coarse-level context:

$$v_{\text{global}} = \lambda_3 \cdot v_{w \rightarrow f} + v, \quad t_{\text{global}} = \alpha_3 \cdot s_{f \rightarrow w} + s, \quad (3)$$

where $v_{w \rightarrow f}$ uses *words as queries over frames as keys/values* (word→frame) to emphasize salient temporal segments, and $s_{f \rightarrow w}$ uses *frames as queries over words as keys/values* (frame→word) to strengthen sentence representations with global visual context.

Final Alignment Between Enhanced Visual and Textual Features.

We fuse multi-level cues into compact modality-specific pattern features that supervise stage II:

$$v_{\text{final}} = \lambda_1 \cdot p + \lambda_2 \cdot f_{w \rightarrow p} + \lambda_3 \cdot v_{w \rightarrow f}, \quad (4)$$

$$t_{\text{final}} = \alpha_1 \cdot w + \alpha_2 \cdot s_{p \rightarrow w} + \alpha_3 \cdot s_{f \rightarrow w}, \quad (5)$$

where $\lambda_1, \lambda_2, \lambda_3$ and $\alpha_1, \alpha_2, \alpha_3$ are learnable weights. Intuitively, v_{final} aggregates patch-, frame-, and video-guided interactions, and t_{final} aggregates word- and sentence-level interactions informed by visual evidence. We treat the well pre-trained $\{v_{\text{final}}, t_{\text{final}}\}$ as the **pattern labels** for distillation in stage II, transferring cross-modal interaction patterns to single-modality encoders.

3.3 Stage II: Label Pattern Distillation for Cross Feature Modeling

As shown in stage II of Fig. 1, we introduce a Cross-Modality Adaptor (CMA) after the CLIP encoders to refine unimodal representations. Formally, given the original sentence embedding s and video embedding v extracted by CLIP encoders, CMA performs a nonlinear mapping:

$$s_{re} = s + ws', \quad v_{re} = v + wv', \quad (6)$$

where $s' = \text{CMA}_t(s)$ and $v' = \text{CMA}_v(v)$ denote the enhanced features for retrieval in a semantically enriched space. This transformation is supervised by the cross-modal interaction patterns pre-trained in stage I, which serve as soft labels to guide the adaptor. In this way, the raw features are encouraged to implicitly attend to modality-specific details revealed by the other modality, rather than relying solely on unimodal evidence.

The key motivation of CMA is that, in retrieval scenarios, paired text–video inputs cannot be used for cross-attention fusion without causing data leakage. To address this, CMA aligns unimodal features with the stronger cross-attention features from stage I, enabling them to inherit complementary semantic cues. This not only reduces the modality gap but also produces more expressive and semantically aligned representations for retrieval.

3.4 Custom Modules for Feature Enhancement

To overcome the limitations of shallow fusion mechanisms that struggle with complex inter-dependencies, we introduce the Dual Attentive Module (DAM), illustrated in Fig. 2. The DAM enables effective cross-modal interaction beyond simple feature aggregation by: (1) *enhancing* the target modality through selective incorporation of salient source context, and (2) *distilling* source features into compact, relevant representations, thus reducing redundancy. This cascaded attention strategy supports semantic enhancement across multiple levels of granularity.

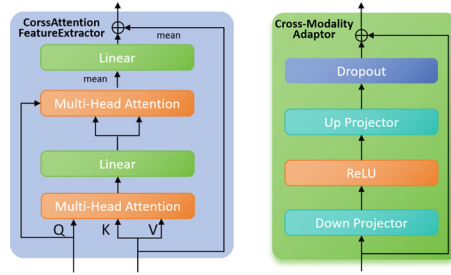


Fig. 2. Custom Modules for Feature Enhancement: DAM and CMA.

Dual Attentive Module. DAM employs a dual-layer residual attention mechanism. Given a query $\mathbf{Q}$ from the target modality, and key $\mathbf{K}$ and value $\mathbf{V}$ from the source, the first attention extracts broad context, which is then refined by a second pass using the original query. Formally:

$$\mathbf{F}_{int} = \text{Linear}_1(\text{MHA}_1(\mathbf{Q}, \mathbf{K}, \mathbf{V})) \quad (7)$$

$$\text{DAM}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathbf{V} + \text{Linear}_2(\text{mean}(\text{MHA}_2(\mathbf{Q}, \mathbf{F}_{int}, \mathbf{F}_{int}))) \quad (8)$$

where $\mathbf{F}_{int}$ denotes the intermediate features. The output is an enhanced version of $\mathbf{V}$ via a residual connection. Applying DAM symmetrically (text-to-video and video-to-text) yields a robust bidirectional fusion framework.

Cross Modality Adaptor. In stage II, CMA maps original features into interaction-enhanced ones, guided by the patterns pre-trained in stage I. It follows a bottleneck design: input features are projected to a lower-dimensional space, activated non-linearly, then restored, with dropout for regularization. A residual connection fuses the transformed and original features, enriching them with fine-grained cross-modal details. Formally, for input $\mathbf{x}$:

$$\text{CMA}(\mathbf{x}) = \mathbf{x} + \text{Dropout}\left(W_u \sigma(W_d \mathbf{x})\right), \quad (9)$$

where W_d and W_u are the down- and up-projectors, $\sigma(\cdot)$ is ReLU, and Dropout denotes regularization.

3.5 Learning Objective

Cross-Modal Fusion Loss for Stage I. To train our model and align the final feature representations (v_{final} and t_{final}), we employ a symmetric contrastive loss. This objective function, commonly known as InfoNCE, aims to minimize the distance between corresponding (positive) video-text pairs while maximizing the distance to non-corresponding (negative) pairs within a batch.

Given a batch of N video-text pairs, the loss is computed bidirectionally: for video-to-text retrieval ($\mathcal{L}_{v \rightarrow t}$) and text-to-video retrieval ($\mathcal{L}_{t \rightarrow v}$).

The video-to-text contrastive loss for a single video v_i is defined as:

$$\mathcal{L}_{v \rightarrow t}^{(i)} = -\log \frac{\exp(\text{sim}(v_i, t_i)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(v_i, t_j)/\tau)} \quad (10)$$

Symmetrically, the text-to-video contrastive loss for a single text t_i is:

$$\mathcal{L}_{t \rightarrow v}^{(i)} = -\log \frac{\exp(\text{sim}(t_i, v_i)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(t_i, v_j)/\tau)} \quad (11)$$

Here, $\text{sim}(a, b)$ denotes the cosine similarity between feature vectors a and b . The term t_i represents the positive text sample corresponding to video v_i , while t_j (for $j \neq i$) are the negative text samples. Temperature τ is a hyperparameter that scales the similarity scores.

To enforce semantic alignment across multiple levels of feature granularity, we extend the contrastive loss to operate on several hierarchical feature pairs. For each level s in the set $S = \{\text{global, coarse, fine, final}\}$, a bidirectional contrastive loss $\mathcal{L}_s$ is computed between the corresponding visual (v_s) and textual (t_s) features:

$$\mathcal{L}_s = \frac{1}{N} \sum_{i=1}^N \left(\mathcal{L}_{v_s \rightarrow t_s}^{(i)} + \mathcal{L}_{t_s \rightarrow v_s}^{(i)} \right) \quad (12)$$

The total loss $\mathcal{L}_{\text{total}}$ for the batch is the weighted sum of these individual losses, controlled by hyperparameters λ_s :

$$\mathcal{L}_{\text{total}} = \sum_{s \in S} \lambda_s \mathcal{L}_s + \mathcal{L}_{re}, \quad (13)$$

where $\mathcal{L}_{re}$ is the retrieval loss proposed in UCoFiA [24].

By minimizing this comprehensive loss, the model learns to project visual and textual features from multiple hierarchical levels into a shared embedding space. This multi-level supervision ensures that semantically related content from different modalities is located closely together, regardless of the feature granularity.

Pattern Distillation Loss for Stage II. We adopt a cosine similarity loss to align the CMA-transformed features with their corresponding soft labels pre-trained and saved in stage I. Formally, the loss is defined as:

$$\mathcal{L}_{\text{distill}} = 2 - \cos(v', v_{\text{final}}) - \cos(s', t_{\text{final}}) \quad (14)$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{re} + \lambda_{\text{distill}} \mathcal{L}_{\text{distill}}, \quad (15)$$

where v' and s' denote the distilled features produced by CMAs, and v_{final} and t_{final} represent cross-modal attentive features pre-trained in stage I.

4 Experimental Setup

4.1 Datasets and Evaluation Metrics

We evaluate our HCMI method on five popular video-text retrieval datasets:

- MSR-VTT [27] contains 10,000 videos, each annotated with 20 text captions. 9,000 videos are used for training, and each video is trained with all 20 captions. Results are reported on the remaining 1,000 videos using the selected caption set (the JSFusion [28] test set).
- DiDeMo [2] contains 10,000 videos and 40,000 captions. All captions of a given video are concatenated into a single query for both training and testing. The training, validation and test set contain 8,395, 1,065 and 1,004 videos, respectively.
- VATEX [23] contains 34,991 videos. We use HGR's [5] split, which uses 25,991 videos for training, 1,500 and 1,500 for validation and testing. This dataset is evaluated with the multiple caption evaluation setting following [24].
- MSVD [3] contains 1,970 videos. The training, validation and test sets contain 1,200, 100 and 670 videos, respectively. This dataset is evaluated with the multiple caption evaluation setting.
- Activity-Net [10] contains 20,000 videos and 100,000 captions. All captions of a given video are concatenated into a single query for both training and testing. The training, validation and test sets contain 10,000, 5,000 and 5,000 videos, respectively.

All videos are compressed to 3 fps and resized to 224 pixels in width or height to speed up the training and evaluation process, according to [16].

We use standard retrieval metrics to evaluate retrieval performance, including recall at rank K ($R@K$) with $K=1,5,10$, median rank (MdR) and mean rank (MnR).

4.2 Implementation Details

Our baseline method is UCoFiA [24]. Following that, we use the pre-trained CLIP ViT-B/32 [19] weights to initialize the visual encoder of our HCMI model. The model architecture and dataset settings are the same as in UCoFiA.

During the first stage, the number of attention heads is set to 8. All λ_s for $s \in S$ in $\mathcal{L}_{\text{total}}$ are set to 0.1, and the temperature τ is also set to 0.1. During the second stage, the down projector projects the feature dimension from 512 to 32, and the up projector projects the hidden state from 32 to 512. Both λ_{distill} and w are set to 1.0 for all datasets except for MSR-VTT, where w is set to 2.0. During both stages, the learning rate is set to $1e-7$ for the visual and text encoder and $1e-4$ for other models. For the first stage, all datasets are trained for 5 epochs, and for the second stage, the number of epochs are 5, 5, 10, 15, 20 for MSVD, VATEX, DiDeMo, MSR-VTT, and Activity-Net datasets, respectively.

5 Experimental Results

In this section, we compare our HCMI method with several prior methods on five video-text retrieval datasets.

5.1 Comparison to State-of-the-Art Methods

MSR-VTT Dataset. As shown in Table 1, HCMI achieves a 1.1% improvement on text-to-video R@1 metric compared to the baseline UCoFiA method and a 0.5% improvement compared to the recent competitor TC-MGC. For text-to-video R@5 and R@10 and video-to-text metrics, HCMI also shows significant improvement over UCoFiA and comparable results to TC-MGC. These results verify that our cross-modality interaction is useful for video-text retrieval.

Table 1. Retrieval results on MSR-VTT. Results of other models came from TC-MGC paper [9]. **Bold** denotes the best and underline denotes the second best.

Method	Text → Video					Video → Text				
	R@1↑	R@5↑	R@10↑	MdR↓	MnR↓	R@1↑	R@5↑	R@10↑	MdR↓	MnR↓
CLIP4Clip [16]	44.5	71.4	81.6	2.0	15.3	42.7	70.9	80.6	2.0	11.6
X-Pool [7]	46.9	72.8	82.2	2.0	14.3	44.4	<u>73.3</u>	84.0	2.0	<u>9.0</u>
CenterCLIP [31]	44.2	71.6	82.1	2.0	15.1	42.8	71.7	82.2	2.0	10.9
DRL [22]	<u>47.5</u>	<u>73.8</u>	<u>83.6</u>	2.0	13.3	46.3	72.7	82.5	2.0	9.5
UCoFiA [24]	46.8	72.6	82.4	2.0	13.9	45.3	72.7	82.2	2.0	9.8
X-CLIP [17]	46.1	73.0	83.1	2.0	13.2	<u>46.8</u>	<u>73.3</u>	84.0	2.0	9.1
TC-MGC [9]	47.4	74.8	84.2	2.0	12.8	45.9	74.5	<u>83.3</u>	2.0	8.6
HCMI (ours)	48.0	72.1	82.5	2.0	<u>13.0</u>	46.9	73.0	82.5	2.0	9.6

DiDeMo and VATEX Datasets. Table 2 shows the results on the DiDeMo and VATEX datasets. In the text-to-video task, HCMI outperforms all compared models, achieving 2.5% and 3.2% improvements in R@1 compared to the second-best TC-MGC results. In the video-to-text task, HCMI also achieves 0.9% - 1.0% improvement in R@1 and delivers comparable result in R@5 and R@10.

MSVD and Activity-Net Datasets. Table 3 shows the text-to-video results on the MSVD and the Activity-Net datasets. The results of the UCoFiA method are obtained by re-training using their source code. HCMI achieves a significant improvement on the Activity-Net dataset and a modest improvement on the MSVD dataset, compared to the baseline UCoFiA method.

Table 2. Retrieval results on DiDeMo and VATEX. Results of other models came from TC-MGC paper [9]. **Bold** denotes the best and underline denotes the second best.

Method		Text $\rightarrow$ Video					Video $\rightarrow$ Text				
		R@1 $\uparrow$	R@5 $\uparrow$	R@10 $\uparrow$	MdR $\downarrow$	MnR $\downarrow$	R@1 $\uparrow$	R@5 $\uparrow$	R@10 $\uparrow$	MdR $\downarrow$	MnR $\downarrow$
DiDeMo	CLIP4Clip [16]	40.8	67.8	77.4	2.0	20.8	40.7	67.3	77.5	2.0	15.4
	TS2-Net [15]	42.5	69.3	77.8	2.0	18.8	42.2	68.2	78.6	2.0	13.6
	UCoFiA [24]	45.2	72.4	81.3	2.0	<u>13.9</u>	44.8	72.0	81.5	2.0	10.7
	X-CLIP [17]	44.7	<u>72.7</u>	80.6	2.0	15.9	44.5	72.3	82.0	2.0	12.6
	TC-MGC [9]	<u>45.7</u>	<u>72.7</u>	<u>81.9</u>	2.0	15.0	<u>45.4</u>	<u>73.3</u>	82.9	2.0	<u>10.5</u>
	HCMI (ours)	48.2	75.4	83.6	2.0	13.7	46.3	73.4	<u>82.7</u>	2.0	10.1
VATEX	CLIP4Clip [16]	58.7	89.3	94.7	1.0	3.7	74.8	96.7	98.6	1.0	<u>1.8</u>
	TS2-Net [15]	59.5	<u>89.8</u>	<u>95.0</u>	1.0	<u>3.6</u>	74.6	96.3	<u>98.9</u>	1.0	<u>1.8</u>
	UCoFiA [24]	59.3	88.8	94.3	1.0	4.1	73.0	96.1	<u>98.9</u>	1.0	1.9
	X-CLIP [17]	59.1	88.9	94.2	1.0	3.9	74.8	97.3	99.0	1.0	<u>1.8</u>
	TC-MGC [9]	<u>60.0</u>	89.5	94.9	1.0	3.8	<u>75.3</u>	97.0	98.6	1.0	1.9
	HCMI (ours)	63.2	91.2	95.7	1.0	3.2	76.3	<u>97.2</u>	98.7	1.0	1.7

Table 3. Retrieval results on MSVD and Activity-Net. $\dagger$ denotes re-training. **Bold** denotes the best.

Method	MSVD					Activity-Net				
	R@1 $\uparrow$	R@5 $\uparrow$	R@10 $\uparrow$	MdR $\downarrow$	MnR $\downarrow$	R@1 $\uparrow$	R@5 $\uparrow$	R@10 $\uparrow$	MdR $\downarrow$	MnR $\downarrow$
CLIP4Clip [16]	45.2	75.5	84.3	2.0	10.3	40.5	72.4	98.2	2.0	7.5
TS2-Net [15]	-	-	-	-	-	41.0	73.6	84.5	2.0	8.4
UCoFiA † [24]	45.7	75.6	84.6	2.0	10.4	42.3	73.2	84.0	2.0	7.4
HCMI (ours)	45.8	76.7	85.6	2.0	9.9	45.3	75.7	85.8	2.0	6.9

5.2 Hyperparameter Tuning and Ablation Study

We conduct an ablation study on DiDeMo to analyze the effect of our method.

Hyperparameter Tuning of the Enhanced Feature Factor w . As shown in Eq. (6), w controls the balance between original CLIP features and CMA-enhanced features. Table 4 shows that text-to-video retrieval performance first improves with larger w and peaks at $w = 1.0$, where the best R@1 and MnR are obtained. This suggests that moderate enhancement is most beneficial, while excessive weighting may suppress useful original features.

Ablation of Using Different Enhanced Features. We conduct an ablation study to analyze the effectiveness of the proposed enhanced features, as shown in Table 5. The Baseline represents our full proposed method with all features

Table 4. Comparison of different hyper parameter w . **Bold** denotes the best.

w	Text $\rightarrow$ Video					Video $\rightarrow$ Text				
	R@1 $\uparrow$	R@5 $\uparrow$	R@10 $\uparrow$	MdR $\downarrow$	MnR $\downarrow$	R@1 $\uparrow$	R@5 $\uparrow$	R@10 $\uparrow$	MdR $\downarrow$	MnR $\downarrow$
0.1	46.5	74.5	84.0	2.0	14.6	43.9	72.3	81.4	2.0	10.3
0.3	47.2	75.2	83.7	2.0	14.0	46.1	73.4	83.0	2.0	10.2
0.5	47.3	75.1	83.4	2.0	13.9	45.7	73.2	83.3	2.0	10.1
1.0	48.2	75.4	83.6	2.0	13.7	46.3	73.4	82.7	2.0	10.1
2.0	47.4	75.2	84.1	2.0	13.9	47.1	73.4	82.7	2.0	10.3
5.0	47.9	74.8	84.0	2.0	14.1	46.6	73.7	82.9	2.0	10.5

enabled. We observed a performance degradation across all ablated components. The w/o $\mathcal{L}_{\text{distill}}$ variant, which removes the distillation loss and its associated pre-trained feature alignment from stage I, shows the most significant drop in performance. This highlights the crucial role of stage I pre-trained features in improving the overall performance of our method. When analyzing the individual contributions of the enhanced features, the w/o v' and w/o s' configurations also show clear declines in performance, confirming that both the enhanced visual and enhanced textual features are essential to the success of our framework.

Table 5. Comparison of different experiment settings

Method	Text $\rightarrow$ Video					Video $\rightarrow$ Text				
	R@1 $\uparrow$	R@5 $\uparrow$	R@10 $\uparrow$	MdR $\downarrow$	MnR $\downarrow$	R@1 $\uparrow$	R@5 $\uparrow$	R@10 $\uparrow$	MdR $\downarrow$	MnR $\downarrow$
Baseline	48.2	75.4	83.6	2.0	13.7	46.3	73.4	82.7	2.0	10.1
w/o $\mathcal{L}_{\text{distill}}$	43.2	69.5	79.7	2.0	16.2	41.7	69.3	79.9	2.0	11.8
w/o v'	46.0	72.3	81.4	2.0	15.9	45.7	72.3	81.6	2.0	12.2
w/o s'	46.5	74.1	82.8	2.0	14.0	44.7	72.1	81.3	2.0	10.4

6 Conclusion

In this paper, we introduce the HCMI framework to overcome the limitations of isolated feature extraction and suboptimal multi-granularity alignment in existing video-text retrieval approaches. HCMI enhances feature representations through multi-level, bidirectional cross-modal interactions during training, while requiring only single-modality input at inference. Extensive experiments show that our framework achieves state-of-the-art performance across multiple benchmarks, striking an effective balance between retrieval accuracy and efficiency. In the future, we plan to extend HCMI to more diverse modalities and investigate its potential in large-scale, real-world retrieval scenarios.

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